# Type MS · Define diffusion, osmosis and active transport

## Answer 1 of 9 · Question 2 part b · 3 marks

2021 NOV · Paper 01 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) | • movement of molecules;<br>• from high to low concentration/down a diffusion/concentration gradient; | | 2marks |
| (b) | any three of<br>• movement of molecules against a concentration gradient/low to high concentration;<br>• using energy;<br>• in the form of ATP;<br>• ref protein channels; | | 3marks |
| (c) | any two of<br>• movement of water;<br>• from high to low water potential/down a water potential gradient;<br>• across a partially/selectively/semi permeable membrane; | ignore ref to high/low conc of water | 2marks |

Total 7 marks

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Define diffusion, osmosis and active transport

Answer 2 of 9 · Question 5 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| **5(a)** | movement of a substance against a concentration gradient/low to high concentration;<br>using energy/ATP; | | **(2)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **5(b)** | • cut pieces of potato to same size/length/mass;<br>• weigh each piece;<br>• place into different solutions;<br>• leave for at least 30 minutes;<br>• remove and dry;<br>• re-weigh/re-measure length;<br>• determine which concentration has no / least change in mass/length; | | **(5)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **5(c)(i)** | 0.10 × 100;<br>7.90<br>1.3; | Ecf allow correct answer to incorrect figures used in calculation<br><br>Allow 1.26 / 1.27 for final correct answer | **(2)** |

## Type MS · Define diffusion, osmosis and active transport

### Answer 2 of 9 · Question 5 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **5(c)(ii)** | initial mass/volume/size varies;<br>greater change in mass/volume if potato is bigger;<br>provides a standardised result;<br>allows comparison; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **5(c)(iii)** | • plot a graph;<br>• concentration vs percentage change;<br>• find (from graph) concentration at which percentage change is zero/lowest change;<br>• this is the concentration of the contents; | | **(4)** |

**Total 15 Marks**

## Type MS · Define diffusion, osmosis and active transport

### Answer 2 of 9 · Question 5 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

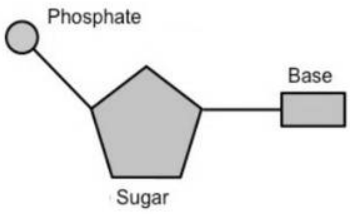
**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · Define diffusion, osmosis and active transport

Answer 3 of 9 · Question 8 part a.i · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 8 (a) (i) | Any two from:<br>• (movement of substances/named solute or gas) against a concentration gradient(1)<br>• using energy/ATP(1) | | 2 |
| (ii) | Any two from:<br>• parent homozygous recessive/heterozygous / carriers (1)<br>• one (recessive) allele inherited from each parent (1)<br>• reference to homozygous recessive disorder/caused by recessive alleles (1) | | 2 |
| (iii) | • protein/CFTR is not modified/folded/protein has an incorrect shape(1)<br>• protein/CFTR not transported (to its destination)(1) | | 2 |
| (iv) | 1408 x 3 (1)<br>4224 | Full marks for correct final answer.<br>Max 2 marks ecf for final answer if it is correct from calculation | 2 |
| (v) | • (deoxyribose) sugar(1)<br>• joined to a (nitrogenous) base/named base(1)<br>• phosphate group attached to sugar (1)<br>  | Allow one mark only if all 3 components just listed<br>Fully labelled diagram for full marks | 3 |
| (b) | any three from<br>• deletion(1) | allow description of deletion | 3 |

Type MS · Define diffusion, osmosis and active transport

Answer 3 of 9 · Question 8 part a.i · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| | • of one codon/TTT (1)<br>• phenylalanine missing from protein (1)<br>• shape of final protein changed/different protein formed (1) | | |

**Total for Question 8 = 14 Marks**

## Type MS · Define diffusion, osmosis and active transport

### Answer 4 of 9 · Question 2 part a · 2 marks

2021 NOV · Paper 01 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) | • movement of molecules;<br>• from high to low concentration/down a diffusion/concentration gradient; | | 2marks |
| (b) | any three of<br>• movement of molecules against a concentration gradient/low to high concentration;<br>• using energy;<br>• in the form of ATP;<br>• ref protein channels; | | 3marks |
| (c) | any two of<br>• movement of water;<br>• from high to low water potential/down a water potential gradient;<br>• across a partially/selectively/semi permeable membrane; | ignore ref to high/low conc of water | 2marks |

Total 7 marks

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Define diffusion, osmosis and active transport

### Answer 5 of 9 · Question 2 part c · 2 marks

2021 NOV · Paper 01 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) | • movement of molecules;<br>• from high to low concentration/down a diffusion/concentration gradient; | | 2marks |
| (b) | any three of<br>• movement of molecules against a concentration gradient/low to high concentration;<br>• using energy;<br>• in the form of ATP;<br>• ref protein channels; | | 3marks |
| (c) | any two of<br>• movement of water;<br>• from high to low water potential/down a water potential gradient;<br>• across a partially/selectively/semi permeable membrane; | ignore ref to high/low conc of water | 2marks |

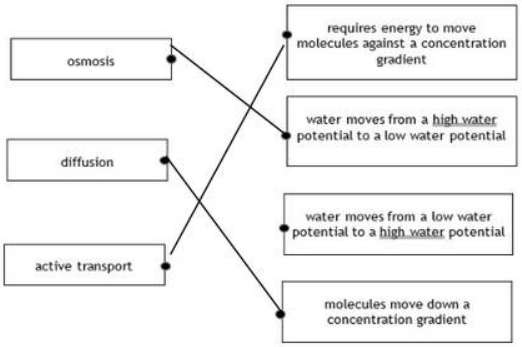
Total 7 marks

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Define diffusion, osmosis and active transport

### Answer 6 of 9 · Question 1 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 1 | (a) (i) | X = cell membrane;<br>Y = nucleus;<br>Z = cytoplasm; | Allow nuclear membrane | 3 |
| | (ii) | mitochondria/ribosomes/endoplasmic reticulum | | 1 |
| | (iii) | • similar cells work together to form tissues;<br>• (different) tissues work together to form organs; | | 2 |
| | (b) |  | Reject more than one line from each process | 3 |
| | (c) (i) | A = squamous (epithelium)<br>B = ciliated (epithelium) | | 2 |
| | (ii) | 60 x 40;<br>2400 (µm) (÷1000);<br>2.4 (mm); | Ecf from mp1. Max 2 marks e.g. 60 ÷ 40 (÷1000) = 0.0015 | 3 |

**Total 14 marks**

## Type MS · Define diffusion, osmosis and active transport

### Answer 6 of 9 · Question 1 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Define diffusion, osmosis and active transport

### Answer 7 of 9 · Question 3 part a.iii · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) | surface area = 24 ($mm^2$);<br>volume = 8 ($mm^3$);<br>3:1; | Full marks for correct final answer | 3 |
| | | (ii) | C 92; | | 1 |
| | | (iii) | movement from a high concentration to a low concentration; | | 1 |
| | | (iv) | • reduces the time taken to become colourless/rate of diffusion/reaction is faster;<br>• the diffusion distance is shorter/greater area over which diffusion/collisions can occur; | | 2 |
| | (b) | (i) | • use acid heated to different temperatures/idea of different temperatures;<br>• compare/measure the time taken to become colourless/compare rate of diffusion; | | 2 |
| | | (ii) | Any two from:<br>• the side length of the cube/size of the cube/surface area/volume of cube;<br>• acid concentration;<br>• concentration of alkali/indicator in the cube; | accept volume of acid<br>Allow volume of indicator | 2 |

Total 11 marks

## Type MS · Define diffusion, osmosis and active transport

### Answer 7 of 9 · Question 3 part a.iii · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Define diffusion, osmosis and active transport

### Answer 8 of 9 · Question 1 part whole · 6 marks

2023 JUN · Paper 01 · find part whole in this mark scheme extract

| | | | |
|---|---|---|---|
| 1 | water;<br>selectively;<br>down;<br>volume;<br>ATP;<br>against; | | 6 |

Total 6 marks

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Define diffusion, osmosis and active transport

### Answer 9 of 9 · Question 7 part b · 2 marks

2024 NOV · Paper 01 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) | • measure initial mass (of food)use same mass of food;<br>• measure initial temperature of water;<br>• burn food;<br>• take final temperature of water;<br>• repeat for each food;<br>• compare temperature difference (of water)/results for each food; | | 6 |
| (b) | • active transport/uptake;<br>• reference to low to high concentration/against concentration gradient; | | 2 |

**Total Marks 8**

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |